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Beyond Human-Centred Design:

Exploring the Complexities of Designing in Entangled Worlds

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Abstract

As digital technologies become more connected to the environment, society, and material world, HCI researchers are called to move beyond anthropocentric perspectives to respond to the complexities of more-than-human design. This thesis proposes a scaffold to support their attunement to the complexities of more-than-human design. Using a qualitative, exploratory approach, this study comprises three phases. In Phase 1, a speculative workshop revealed the challenges HCI researchers face when engaging with more-than-human design. Building on these findings, Phase 2 involved interviews with experts in the field to explore these challenges in greater depth. Finally, in Phase 3, we engaged directly with more-than-human design practices through attunement activities, allowing us to personally experience and reflect on these complexities. Key findings revealed the challenges of designing in entangled worlds, as well as the tools and methods researchers use in more-than-human design. The analysis identifies key qualities of attunement, proposing a scaffold composed of three entangled modes: Wondering, Sensing, and Reflecting. These modes foster curiosity, embodied awareness, and critical reflection, helping researchers cultivate relational mindsets. Rather than prescribing solutions, it encourages HCI researchers to stay with complexity and engage responsibly. This work contributes to more caring, interdependent, and sustainable approaches in the field of HCI as well as design education and practice.

Keywords: More-than-human design, human-computer interaction, design tools, relational thinking, attunement, thoughtful design, speculative design, sustainable design

1. Introduction

As digital technologies become increasingly entangled with the ecological, social, and material worlds, they shape—and are shaped by—complex relationships that are constantly evolving together (Barad, 2007; Frauenberger, 2019; Morrison and McPherson, 2024). The technologies we design and craft are not merely present in the world. They are understood as fluid assemblages—constantly changing interconnections of components, tools, services, platforms, infrastructures, and materials (Redström and Wiltse, 2018). This has also entangled us in webs of increasingly complex interactions (Janlert and Stolterman, 2017). Digital design in today's world is, therefore, increasingly challenging, involving ambiguity, ethical dilemmas, and the task of envisioning and crafting what does not yet exist within various constraints.

This highlights the importance of critical reflection in design practice (Löwgren and Stolterman, 2007; Agre, 2014). Critiques of anthropocentric assumptions in design emphasise the need to embrace a posthumanist perspective in the design process to acknowledge the agency of both human and non-human entities (Wakkary, 2021). The field of More-Than-Human Design (MtHD) has, over the last ten years, evolved as a vibrant field in HCI where relational complexities challenge stable categories/identities. This shift does not intend to devalue humans but rather acknowledges that humans are no longer the only agents in the co-constitution of things, prompting a reframing of the conceptual foundations of design. Accordingly, design is moving beyond linear, one-to-one interactions between human users and artefacts toward practices embedded within complex assemblages of humans,

nonhumans, and technologies (Giaccardi and Redström, 2020; Coşkun et al., 2022; Giaccardi, Redström and Nicenboim, 2024).

Today's HCI research is increasingly entangled with sustainability, ethical inclusion of diverse human and non-human actors, and the ongoing co-constitution of sociotechnical and ecological worlds (Eriksson et al., 2024). HCI research aims to influence design practice; however, existing frameworks tend to be overly scientific and analytical, often failing to address its situational complexity. This has led to the need for studying and theorising design practice within HCI research, requiring deep engagement with designers in their actual work (Goodman, Stolterman and Wakkary, 2011). The field is experiencing a heightened push to move beyond purely theoretical discourse toward actionable, practice-based guidance—especially as designers face urgent, real-world crises that require them to embrace more-than-human, relational modes of thinking (Nicenboim et al., 2024). Despite calls for a shift towards more-than-human and posthuman thinking, designers still struggle to move beyond anthropocentric assumptions and attune to relational modes of thinking (Dörrenbächer et al., 2024). It seems as if the necessary relational thinking demands a different register—one that involves new forms of expertise, theories, and frameworks (Forlano, 2017).

In addition, the foundational elements needed to develop design research and education programmes that explore and ethically reorient designers towards more-than-human practices remain implicit (Giaccardi and Redström, 2020). Despite this growing discourse, we still lack a detailed understanding of the specific challenges HCI researchers face in more-than-human design, which is critical for developing methodologies that bridge theory and practice. Without this knowledge, there is a critical gap in how HCI can support its own community. Understanding these practical realities is essential for developing new methods, tools, and educational pathways that help researchers attune to relational complexity—ensuring that HCI research can truly inform and improve design practice in this entangled, more-than-human era.

This thesis responds to these gaps by focusing on HCI researchers themselves as key actors in bridging theory and practice, aiming to support HCI researchers in engaging with and attune to the complexities of more-than-human design. HCI researchers inform design practice; therefore, understanding the challenges they face/identify in engaging with more-than-human design is crucial for supporting the field's evolution. To explore this, we have adopted a qualitative, exploratory approach, unfolding across three phases—speculative design workshop and individual subsequent interviews, expert interviews, and researcher attunement activities—complemented by a continuous process of reflective journalling and literature engagement.

1.1 Research Question

What challenges do HCI researchers identify in more-than-human design?

1.2 Significance

This thesis is significant because it addresses a critical gap at the intersection of HCI research, design practice, and more-than-human thinking. By exploring the challenges HCI researchers face as they engage with more-than-human design, this work provides grounded, practice-

based knowledge that can inform new methods, tools, and educational pathways. In doing so, it supports HCI researchers in moving beyond anthropocentric assumptions and equips them to attune to relational complexity with greater ethical sensitivity and practical insight. More broadly, this research contributes to the ongoing evolution of design practice by developing actionable guidance for researchers who bridge theory and practice. Its findings offer value not only for HCI researchers and educators but also for the wider design community striving to integrate sustainability, diverse agencies, and more-than-human perspectives into their work. In this way, the thesis advances both scholarly discourse and real-world practice — helping HCI become more responsive to the entangled sociotechnical and ecological worlds that designers must increasingly care for and co-construct.

2. Related Research

This section introduces the field of more-than-human design, tracing its evolution from reflective and critical design practices in HCI to more relational and posthuman perspectives. It also reviews previous research on the role of tools and methods to engage with more-than-human complexities, as well as studies on designerly tools in Human-Computer Interaction and their key characteristics for engaging with complex design contexts.

2.1 Thoughtful Design: Embracing, Reflective and Critical

As design increasingly involves dynamic interactions between people and systems, complexity and uncertainty have become defining features of contemporary practice (Giaccardi and Redström, 2020). Agre (2014) argues that technical practitioners must embrace this difficulty through both craft and critique, echoing Löwgren and Stolterman's (2007) call for "thoughtful designers" who critically reflect on their roles, assumptions, and the broader consequences of their decisions. This demands a shift from technical efficiency to reflective, ethical, and aesthetic engagement—combining intuition, experience, and critical judgment over fixed methods.

Such reflection is now central in design research. Goodwill, Bendor, and van der Bijl-Brouwer (2021) propose a power literacy framework for engaging systemic inequalities, while Vallgårda (2024) shows how care technologies can unintentionally cause harm, stressing ethical awareness. Drawing on feminist theory, Croon (2022) introduces "contact zones" for ethical exploration, and Epp et al. (2024) conceptualise the design process as engaging with both visible and invisible uncertainties—positioning uncertainty as a generative design resource.

In response to issues around sustainability concerns, representation and inclusivity, the increasing agency of technologies, and the entanglements of human and nonhuman bodies in design practice, research about more-than-human design in HCI community is growing (Eriksson et al., 2024). Several paradoxes of more-than-human design have been identified, revealing the complexities and dilemmas that designers must acknowledge, reflect on, and embrace when designing for nonhuman actors (Dörrenbächer et al., 2024). Núñez-Pacheco and Rosén (2024) explored how interactions with nature and technology connect to ethical

tensions, sensory richness, and the dynamic boundaries between self and environment, calling for more attuned, relational approaches in design.

2.2 Relational Entanglements and Systems Thinking

Building on the reflective and critical foundations outlined in the previous section, recent design research has increasingly turned toward relational and posthuman perspectives to better engage with the ethical, ecological, and systemic complexities of design. Posthumanism critiques human-centred assumptions, proposing a more relational understanding of subjectivity and new ethical and ontological frameworks (Braidotti, 2013). Relational ontology further challenges individualism by positioning relationships as the basis of existence (Sidorkin, 2002). These perspectives expose the limitations of anthropocentric design in tackling socio-ecological challenges (Forlano, 2017), calling for a redefinition of what it means to design—and to be human—within entangled systems (Giaccardi, Redström and Nicenboim, 2024).

In HCI, this shift is reflected in frameworks such as Entanglement HCI (Frauenberger, 2019) and more-than-human design, which integrate ecological, multispecies, and posthuman perspectives (Coşkun et al., 2022). Rosén (2022) highlights the role of noticing, sensory engagement, and context-specific technologies in fostering meaningful relationships with nonhuman entities in urban farming. These inherently complex, systemic issues call for holistic approaches such as systems thinking, which emphasizes patterns, relationships, and dynamic interconnections (Capra, 1996; Nelson and Stolterman, 2014). Practising more-than-human design requires approaches that are both action-oriented and critically reflective. To support this, design researchers are developing tools that translate theoretical insights into practical guidance for engaging with relational and systemic complexity (Rosén et al., 2024a).

2.3 More-Than-Human Design

In more-than-human design, a diverse range of tools, methods, and approaches support designers in engaging with complex entanglements from situated, relational perspectives (Rosén et al., 2024a).

Participatory approaches, for instance, extend beyond designing for nonhumans to designing with them. Westerlaken and Sandelin (2024) highlight how participatory design with animals enables their active involvement, informed by Critical Animal Studies which shifts attention from symbolic to lived conditions. Cerna et al. (2024) explore plant relations through three modes of entanglement—collective fabulation, critical companionship, and cohabitation, introducing workshops, naming practices, and IoT-enabled care labs to challenge plant awareness disparity.

Wilde and Lenskjold's (2024) participatory research through design (pRtD) approach engages with the human microbiome using food-based workshops, speculative self-experimentation, and material provocations to cultivate microbial awareness. Similarly, Woytuk and Søndergaard (2024) introduces a rich set of tools and methods for designing with the more-than-human body such as biomaterial prototyping, 3D-printed moulds, and pH-sensitive dyes to connect bodily fluids with material response.

Rosén et al. (2024b) propose reflective and relational tools such as system mapping, philosophical cards (e.g., agential cutters, leverage pointers), and critical questioning prompts to bridge theory and practice. Additional methods include poetic field notes, speculative fiction, embodiment probes, and biological empathy mapping (Laurien et al., 2024; Biggs et al., 2024), all of which aim to support designers in making visible, felt, and accountable their entanglements with more-than-human worlds.

2.4 Scaffolding Design Practices through Tools

Design tools are not passive. They are more than facilitator of action, as they also shape designers' perception, cognition and the understanding of problems and solutions (Dalsgaard, 2017). Rogers (2009) called for new thinking tools that support the early stages of design by addressing complex human values, ethical tensions, and speculative inquiry. As designers address today's problems, they should also focus on developing tools that enhance their knowledge, making it accessible and reusable for others. However, many research-driven tools fail to align with the realities of design work. The concept of designerly tools refers to methods and approaches that support creativity, iteration, and problem-solving in ways valued by practitioners (Stolterman et al., 2009).

Buur et al. (2023) introduced Theory Instruments—tangible tools that help designers engage with theoretical frameworks and reveal hidden system dynamics. Similarly, card-based tools are valued for their flexibility and tangibility; they structure thinking and facilitate idea generation and collaboration without requiring deep domain expertise (Wölfel and Merritt, 2013; Roy and Warren, 2019). While the effectiveness and usability of design tools are crucial considerations in their development, understanding the relationships between designers and their tools also provides valuable insights for improving tool design. Designers' choice of tools is influenced not only by efficiency and usability but also by their habits, personal identity, and disciplinary culture. A deeper understanding of the designer-tool relationship can contribute to design education, professional practices, and tool development (Stolterman and Pierce, 2012).

3. Methodology

In this section, we present our study design, participants, data collection, literature review, data analysis, and research ethics. In order to explore what challenges HCI researchers face/identify in more-than-human design, we have adopted a qualitative, exploratory approach, informed by previous researcher on thoughtful interaction design (Löwgren and Stolterman, 2007; Agre, 2014), posthumanism (Forlano, 2017; Wakkary, 2021), and more-than-human design (Nicemboim et al., 2024). This framing aligns with the growing body of HCI literature calling for relational, reflective, and attuned design practices (Frauenberger, 2019; Croon, 2022).

3.1 Study Design

This study was structured in three phases. In the first phase we conducted a speculative design workshop inspired informed by (Rosén et al. (2024b) and a follow up interview with the participants. The second phase is based on interviews with design researchers doing research

in the field of More than Human Design. The third phase involves our own attunement and sense-making activities. Each phase is further described below.

3.1.1 Phase 1: Speculative Design Workshop and Follow-Up Interviews

The first phase involved a speculative design workshop with four PhD students in HCI. Subsequently, three of the four participants participated in follow-up interviews (Table 1).

Motivated by the Four Questions for Systemic More-Than-Human Design in Practice (Rosén et al., 2024b), a recently published more-than-human design method developed by scholars in the field, the first phase involved a speculative design workshop with four PhD students. Their diverse HCI backgrounds provided a valuable starting point for examining the challenges HCI researchers face/identify in more-than-human design through hands-on design activities. Subsequently, three of the four participants participated in follow-up interviews (Table 1).

Participant	Background	Gender	Participated in
P1	HCI PhD student	Female	Workshop + Interview
P2	HCI PhD student	Female	Workshop + Interview
P3	HCI PhD student	Male	Workshop + Interview
P4	HCI PhD student	Female	Workshop

Table 1 *Participants of Phase 1*

Despite their valuable professional backgrounds, it is important to acknowledge that the sample was limited to a small group of Sweden-based PhD students, which may not fully reflect the diverse realities, challenges, and constraints that professional designers face when engaging with more-than-human complexities.

We selected and integrated several tools to support participants' inspiration and creativity in design process (Stolterman and Pierce, 2012). We adopted speculative design both as a method and as a tool to help participants envision future possibilities (Zhu, Choa and Fu, 2024) and critically engage with the complexities of more-than-human design throughout the workshop activities, which aligns with the calls for speculative and critical design practices that address ethical tensions and support reflection (Löwgren and Stolterman, 2007; Rogers, 2009; Croon, 2022).

Coffee is deeply rooted in Swedish culture (Nelson and Svanberg, 1993) and was used in the workshop to encourage participants to reflect on their daily routines and the broader systems connected to them, as part of exploring more-than-human design. The workshop was structured around the Four Questions for Systemic More-Than-Human Design in Practice (Rosén et al., 2024b), which guided participants in exploring systemic relationships, the selection of entities, mapping practices, and intervening in more-than-human systems.

As part of our preparation for the workshop, we adapted and expanded the prompt materials. To prompt and support participants in engaging with more-than-human thinking throughout the activities, we adopted two sets of prompt cards proposed by Rosén et al. (2024b): Agential Cutter cards and Leverage Pointer cards (Figure 2). The Agential Cutter cards used in the workshop were developed based on a typology of more-than-human entities

in nature identified by Veselova and Gaziulusoy (2021) to assist participants in identifying which entities should be included in their system maps. Since this typology primarily focuses on entities found in natural systems, we expanded the original set by incorporating additional categories—such as human-cultural systems, tools and technologies, and material cycles—to better reflect the broader range of relevant entities and relationships within the system. The Leverage Pointer cards, inspired by Meadows' (1999) concept of leverage points, guided participants in identifying intervention points through design. Each card included a reflective question, a brief explanation, and an example. Due to time constraints, a selected set of seven leverage points was used.

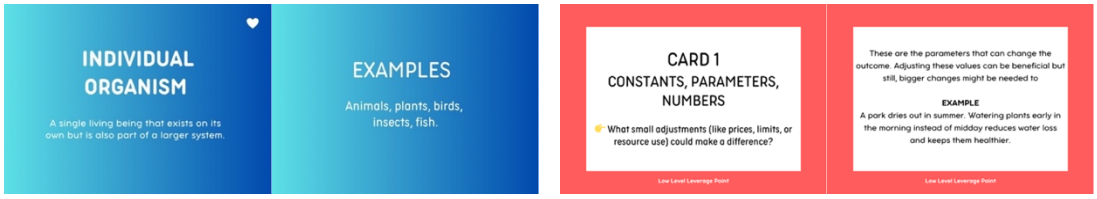


Figure 2 *Example of the Cards (Front and Back Sides)*

We engaged in the workshop as facilitators. Throughout the session, we guided participants through a series of activities, which included sharing their lived experiences with coffee and reflecting on the various human and non-human contributors to those experiences, mapping coffee-system entanglements, designing speculative artefacts or scenarios, and proposing as well as discussing their activities outcomes (Figure 1). Our facilitation aimed to support participants in exploring more-than-human design and to encourage deeper reflection on more-than-human connections across each stage of the workshop.

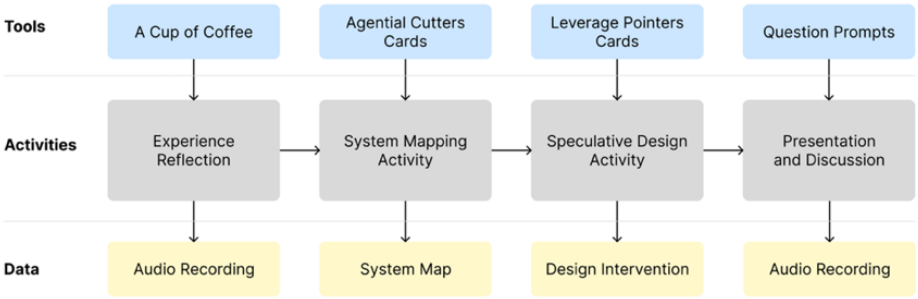


Figure 1 *Speculative Design Workshop Structure*

After the workshop, follow-up interviews with three participants explored their experiences engaging with more-than-human design, with a particular focus on the tools used, their understanding of the system, and their intervention choices. The findings revealed that participants struggled to fully engage with the workshop activities. Discussion during the workshop and responses from the interviews highlighted that the materials—especially the Agential Cutter and Leverage Pointer cards—did not effectively support participants in opening up to more-than-human design thinking. All four participants responded that they found the cards unhelpful, especially the Leverage Pointer cards, which they described as too abstract and containing too much information. Instead of prompting exploration, the cards often became a barrier to design, discussion and reflection. This phase revealed key challenges faced by HCI PhD student, expressed as design paradoxes.

3.1.2 Phase 2: Interviews with More-Than-Human Design Scholars

Because the participants in Phase 1 faced many challenges when engaging with more-than-human design during the workshop, we chose to focus Phase 2 on interviewing scholars with more experience in the field of more-than-human design. The second phase involved semi-structured interviews with five scholars who specialise in more-than-human design. Among these participants, two identified as female and three as males. They were recruited for their deeper knowledge of relational thinking and experience working within systems that include both human and nonhuman entities. To protect participants' identities, all participants were assigned ***pseudonyms***, which are used throughout the thesis (Table 2).

Name	Background	Gender	Focus Area
Tessa	Design	Female	Embodied and co-creative methods for sustainable transformation
Elias	HCI	Male	Sustainable futures and more-than-human perspectives
Alex	Industrial design	Male	Multi-species collaboration within ecological systems
Casey	Design	Female	Entanglement of digital technologies with bodies and ecologies
Linus	Design	Male	Speculative, posthumanist approaches to material culture and emerging technologies

Table 2 *Participants of Phase 2*

The purpose of these interviews was to gather deeper insights into the challenges of more-than-human perspectives in their work. The interview questions were informed by the results from the workshop and its follow-up interviews in the first phase. The interview questions aimed to explore the challenges researchers and designers face when engaging with nonhuman perspectives, and the tools, methods, or practices they use to attune themselves—and others—to more-than-human ways of thinking and designing in different stages of the design process.

3.1.3 Phase 3: Our Attunement Activity

This phase was motivated by the need to engage with more-than-human design through thoughtful practice, intra-actions, and situated embodiment (Löwgren and Stolterman, 2007; Croon, 2022; Núñez-Pacheco and Rosén, 2024). We undertook context-specific, embodied activity to attune ourselves to more-than-human complexities in the everyday life. These reflections served as tools for reflexive synthesis, grounding abstract insights in lived experiences and supporting the transition into analysis and discussion. They also helped bridge earlier findings with personal perspectives, deepening our understanding of the challenges HCI researchers face/identify in more-than-human design.

3.1.4 Reflective Journal

To cultivate our own judgment and engage with the research space in a more critical and reflective manner (Löwgren and Stolterman, 2007), we aimed to embed reflexivity throughout the research process by maintaining a shared reflective journal. The journal allowed us to document and critically engage with our evolving thoughts, emotional responses, emerging questions, and the tensions faced during the research activities, from data collection to analysis. Entries were made individually and discussed collectively at various points in the

process. These reflections became a space for sharing thoughts, exploring our own assumptions, attending to discomfort and uncertainties, and capturing shifts in our perspectives as the project progressed.

3.2 Data Collection and Literature Review

Data collection was conducted in three phases, aligned with the research question of exploring what challenges HCI researchers identify in more-than-human design. Each activity involved different data collection methods tailored to its specific focus (Table 3). Data types included audio recordings, visual artefacts (such as system maps and speculative designs), written reflections, and field notes. All interviews were audio recorded and later transcribed for analysis.

Methods	Who Was Involved	Duration	Data Types
Speculative design workshop	4 HCI PhD students	150 minutes	Audio recordings, visual data of speculative design and system maps, field notes
Semi-structured interviews	3 participants from the workshop	~30 minutes each	Audio recordings, field notes
Semi-structured interviews	5 experienced scholars in more-than-human design	~45 minutes each	Audio recordings, field notes
Attunement practices	2 authors	Varied	Written, conceptual visualisations
Reflective journal	2 authors	Throughout the study	Written

Table 3 *Data Collection Structure*

3.2.1 Literature Review

To inform this thesis, a literature search and iterative reading process were conducted, forming the conceptual, methodological, and analytical foundation of the study. Relevant works were identified through academic databases (e.g., ACM Digital Library, Google Scholar, DiVA), curated reading lists, and reference tracing. Keywords included “more-than-human design,” “posthumanist HCI,” “sustainable HCI,” “attunement,” and “relational thinking.” The literature spans theoretical (e.g., Capra 1997; Braidotti 2013), HCI-focused (e.g., Wakkary 2021; Frauenberger 2019), and practice-based contributions (e.g., Buur 2023; Cerna et al. 2024).

Reading is also an ongoing method of sensitisation and critical attunement. This aligned with perspectives in feminist and posthumanist design (e.g., Croon 2022; Forlano, 2017), where reading is understood as an act of inquiry and situated knowledge production. Through repeated engagement with literature, themes of more-than-human entanglement, relationalities, complexities, and sustainability were later used to shape the empirical design and analytical lenses of this study. Selected works illustrate the growing shift in HCI toward more-than-human perspectives and the need for new scaffolds to support such transitions (Table 4).

Source	Contribution/Focus	Insights
Frauenberger (2019)	Proposal of Entanglement HCI as a new paradigm	• Challenges traditional HCI paradigms by highlighting socio-technical entanglements.
Giaccardi and Redström (2020)	Critique of human-centred design in digital contexts	• Limitations of HCD, shift toward relational • Need for new design practices embracing uncertainty • Technology as active participant
Eriksson et al. (2024)	Scoping review of more-than-human research in HCI	• The motivations in more-than-human design research in HCI including: sustainability (50%), representation (23%), technology as actor (15%), and human-nonhuman body relations (12%)
Dörrenbächer et al. (2024)	Paradoxes and challenges in designing for nonhumans	• Challenges: anthropocentrism and relational thinking • Embrace paradoxes rather than resolve them • Designers' ethical responsibility through response-ability
Giaccardi, Redström and Nicenboim (2024)	Editorial framing for more-than-human HCI special issue	• Design as transformation of the self • Ethical necessity of MTHD in facing planetary crises • Complexity as a generative design space
Vallgårda (2024)	Care technologies and attunement	• Care as relational, vulnerable, and responsive • Attunement as co-becoming

Table 4 *Literary Contributions Framing the Need for More-Than-Human Design in HCI*

3.3 Data Analysis

The analysis is shaped by our ambition to develop a scaffold intended to support HCI researchers to attune to the complexities of more-than-human design. The term "scaffold" is used as metaphor to indicate the supportive flexible and temporary nature of the modes and the prompts that can help HCI researchers to engage with the complexities in more than human design (Wood, David and Ross, 1975; Verenikina, 2003).

Given the exploratory nature of the study, an inductive thematic analysis was conducted to analyse the data (Braun and Clarke, 2006). Audio recordings from the workshops and interviews were transcribed and repeatedly reviewed. Data from Phase 1 and Phase 2 were analysed, alongside insights with our attunement activities and reflective journalling, which served as reflexive lenses. Subsequently, manual coding was performed to identify themes related to the challenges that HCI researchers face/identify in more-than-human design. Additional data, including participant-generated materials from the workshop, observation notes, and attunement outputs, were also analysed.

In Phase 1, 54 initial codes were identified and clustered into 11 key themes: conceptual abstraction, misunderstandings of the concept, low-perceived relevance, uncertainty about the consequences of interventions, lack of domain knowledge or perspective gaps, empathy gaps, limitations stemming from closed-lived experience, cognitive overload, cultural frictions, tool ambiguity and overwhelm, and challenges in understanding the tools. These key themes were then further grouped into broader main themes that describe the primary challenges faced by HCI researchers when engaging with more-than-human design during the workshop (Table 5)

Challenges Faced by Participants During Phase 1 Workshop										
Concept Unfamiliarity				Aware and Unaware Anthropocentrism			Tools-Related Challenges			
Conceptual Abstraction	Misunderstanding of the Concept	Low Perceived Relevance	Uncertainty in Consequences	Domain Knowledge / Perspective Gaps	Empathy Gaps	Closed Lived experience Limitation	Cognitive Overload	Cultural and Personal Frictions	Tool Ambiguity and Overwhelm	Understanding the Tools
Initial Codes: 54 codes										

Table 5 *Themes Emerged from Phase 1*

In Phase 2, 120 initial codes were identified and clustered into 17 themes including irrelevance and complexities, academic and practice gaps, decentring human, speak for other beings, relationality, wonder, reflexivity and positionality, content specific, situatedness and temporal ethics, immerse to sensitise, cohabiting and ethnography, participatory and co-creation, embodiedand sensory engagement, hands on making and emergence acceptance as a way to sensitise, storytelling and questions, change beyond behaviour, and personal process. These themes revealed both challenges expert HCI researchers faced when engaging with more-than-human design. These key themes were then further grouped into broader main themes that describe the inherent complexities of more-than-human design and tools and methods used to engage with such complexities (Table 6).

Challenges and Methods faced/identified by Experts																
Inherent Complexities										Tools and Methods						
Relational Entanglements		Empathy Gaps			Power Relations		Uncertainties, Situatedness			Sensory and Embodied Attunement		Participatory Engagements			Embracing Experiments /Emergences	
Challenges		Approaches								Tools and methods						
Irrelevance and complexities	Academic and practice gaps	Decentring human	Speak for other beings	Relationality	Wonder	Reflexivity / positionality	Context specific	Situatedness and temporal ethics	Immerse to sensitise	Cohabiting / ethnography	Participatory and Co-creation	Embodied and sensory	Hands on making / emergence	Storytelling / questions	Change beyond behaviour	Personal process
Initial Codes: 120 codes																

Table 6 *Themes Emerged from Phase 2*

Through ongoing analysis and reflection across Phase 3 and our shared reflective journal, we identified several challenges that emerged during our engagement across all phases — from facilitating the workshop, to conducting interviews, to participating in attunement activities. These include: the persistence of anthropocentrism, the difficulty of attuning to the invisible, and the potential of questions as tools for rethinking. We also identified key qualities that can support attunement: wondering rather than following prescriptions; decentring the human; multisensory and embodied engagement; embracing ambiguity and situatedness; and fostering reflexivity and positional awareness.

To make these qualities more accessible for HCI researchers, we aim to build a scaffold that supports them in engaging with the challenges of more-than-human design. Building on embodied insights from multiple phases of data collection, we iteratively mapped the meaningful themes that emerged from the qualities of attunement—derived from the challenges identified by participants—to define the core elements of a scaffold to support HCI researchers in engaging with the complexities of more-than-human design. Our findings highlight attunement as a personal, developmental process that requires different modes of thinking. From the themes, we conceptualised three core modes that structure the scaffold: a mode of wonder (encouraging wondering rather than following prescriptions, embracing ambiguity, and remaining situated); a mode of reflection (decentring the human, and fostering reflexivity and positional awareness); and a mode of sensing (engaging the multisensory and

embodied aspects of design practice). Together, these modes offer a framework to help researchers deepen their attunement to more-than-human relationalities.

3.4 Ethical Considerations

Ethical considerations were therefore central throughout all phases of this study, ensuring transparency and respect for participants. All participants were provided with informed consent in accordance with GDPR and ethical research guidelines. The consent forms outlined the purpose of the study, the activities involved, data protection, potential risks and benefits, and participants' rights regarding voluntary participation and withdrawal.

All procedures followed the Swedish Research Council's four principles of good research practice (Vetenskapsrådet, 2024), particularly in cases involving potentially sensitive or personal experiences. This work also builds on existing HCI literature that addresses the ethical dimensions of conducting research in contexts where marginalisation or relational vulnerability is present. These ethical commitments are not only procedural but also reflect the ethos and methods of the design approach taken in this thesis.

3.5 Methodological Reflections and Limitations

Throughout the study, we continuously reflected on the methodological choices, limitations, and situated nature of our research process. As a qualitative, exploratory study, our research was shaped by the tools, contexts, and positionalities of both participants and researchers.

In Phase 1, we used several tools in the speculative design workshop, including coffee, the Four Questions for Systemic More-Than-Human Design in Practice (Rosén et al., 2024b), Agential Cutters cards, and Leverage Pointer cards. We acknowledge that using coffee as a relational medium had its limitations, as participants' cultural backgrounds and personal preferences may have influenced their engagement with the activities. While the Four Questions for Systemic More-Than-Human Design in Practice (Rosén et al., 2024b) provide a useful guideline for systemic more-than-human design, they risk they risked oversimplifying complex entanglements and reinforcing anthropocentric assumptions. The Leverage Pointer cards, inspired by Meadows' (1999) concept of leverage points, were limited to a selected set of seven due to time constraints, may have narrowed the systemic scope explored.

While the speculative design workshop effectively provoked imaginative and relational thinking, it lacked ongoing, embodied engagement with real-world more-than-human entanglements. Its short duration limited the development of deeper, situated understanding and attunement. The paradoxes identified are based solely on this context and may not capture the full range of tensions in more-than-human design.

In Phase 2, the interviews with the participants as a data collection method relied on participants' reflections and words rather than direct observation of their practices, potentially introducing subjectivity and interpretation biases. Moreover, four of the participants were based in Sweden and one in the Netherlands, meaning that the insights may be shaped by a Northern European cultural context, potentially limiting their applicability to other geographical or cultural settings.

In Phase 3, our attunement practices were conducted under time constraints, limiting deeper engagement with relational complexities. These activities were individually defined and

context-specific, which allowed for personal relevance and embodied engagement, but also introduced variability and subjectivity.

We maintained a reflective journal throughout the process. As a self-reported and internally discussed method, the journal offered interpretive depth but was shaped by our positionalities and lacked external validation. As workshop facilitators and interviewers—we were active participants rather than detached observers, and our situated perspectives inevitably influenced the analysis. Rather than aiming for neutrality, we embraced reflexive journaling to critically examine our assumptions and highlight how our engagement shaped the insights presented.

Furthermore, our proposed tentative scaffold is interpretive and exploratory. It was developed through thematic analysis but has not yet been validated through diverse real-world applications. It should be viewed as an invitation for further critique, refinement, and adaptation, rather than a definitive framework.

4. Results

The empirical material collected throughout the study is presented in this section, aiming to illustrate the key findings that inform the research question: What challenges do HCI researchers faced/identify in more-than-human design?" Emerging themes include participants' struggles with the unfamiliar and various approaches to engaging with more-than-human entanglements. These insights inform the development of a scaffold to support HCI researchers in attuning to the complexities of more-than-human design.

4.1 Challenges faced in Speculative Design Workshop

In Phase 1, participants created system maps and proposed speculative interventions (Figure 3), revealing key challenges in engaging with more-than-human design. Many struggled with its abstract nature and expressed confusion around tools like the Agential Cutters cards. Rather than approaching the exercises openly, participants often adopted a structured, task-oriented mindset, focusing on systematically answering the questions.

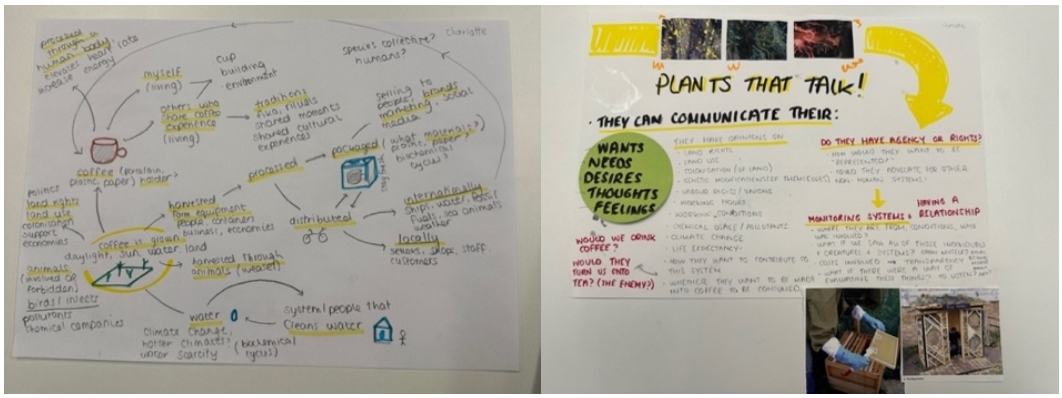


Figure 3 *Example of Workshop Outcomes*

The empirical materials points to the struggles participants face when engaging with more-than-human perspectives in design. These findings informed the underlying challenges that

contribute to HCI researchers' struggles with more-than-human thinking, including conceptual unfamiliarity, aware and unaware anthropocentrism, and tool-related challenges.

4.1.1 Concept Unfamiliarity

Many participants struggled to understand the foundational concept of more-than-human design, finding it abstract and difficult to relate to. This was evident in the workshop, follow-up interviews, and expert accounts, where HCI researchers questioned its relevance to design practice. The unfamiliarity caused discomfort when using tools like the Agential Cutters and Leverage Pointers, highlighting broader challenges in adopting radically different design paradigms. Participants often defaulted to human-centred concerns, especially around labour, consumerism, and social justice. One participant, in particular, critiqued more-than-human design as distant and impractical, stressing that meaningful change should start with addressing the needs of marginalised human communities:

> *"My critique of more than human design is that it's very abstract... I always struggle when I read papers on more-than-human design... where is the real-world relevance? ... I always think it starts with humans and sort of ends with humans, because unfortunately we have that power... Sorry, not sorry. Humans first" (p3).*

This perspective illustrates a critical challenge to more-than-human design: without grounded, accessible entry points, the conceptual unfamiliarity may not only inhibit engagement but also trigger resistance. For some practitioners, the framing of more-than-human perspectives may appear misaligned with urgent ethical priorities.

4.1.2 Aware and Unaware Anthropocentrism

One of the most apparent struggles was still the difficulty of moving beyond anthropocentric assumptions. In the speculative design workshop, some speculative interventions relied on anthropomorphism as a mean to bridge the gap between the researchers and nonhuman entities. While anthropomorphism can serve as an entry point, it often results in the imposition of human norms and values, rather than genuinely accounting for nonhuman perspectives on their own terms. The responses include:

> *"[I] created plants that can talk, and they can communicate with us... they have opinions... thoughts about colonisation... labour... life expectancy... whether they want to be made into coffee..." (p1),*

> *"I'm generally thinking about human experience... it's a little harder to empathise with non-humans, if that makes sense" (p3).*

4.1.3 Tools-Related Challenges

The tools used in the workshop—such as coffee, Agential Cutters cards, system mapping activities, and Leverage Pointers cards—were intended to provoke relational thinking. However, they were often perceived as abstract, overwhelming, and difficult to interpret. This

suggests a misalignment between the goals of the tools and the cognitive, cultural, and paradigmatic readiness of the participants. In some cases, the fixed nature of the tools may have reinforced resistance and limitations rather than supporting openness and exploration. The responses include:

> *"The cards were very difficult in the beginning because they are very abstract" (p4),*
>
> *"They [the cards] did not do much for me… they were very abstract" (p2),*
>
> *"There's so many [cards], and it's so much to think about… and then I come back to the kind of mental effort that you need to put in" (p1).*

4.2 Emergence of Paradoxes

The speculative design workshop and its follow-up interviews (Phase 1) revealed some paradoxes that highlight how well-intentioned interventions can unintentionally lead to contradictions and dilemmas (Table 7).

Paradox	Description	Tension
Supporting Sustainable Coexistence by Erasing Relationships	A speculative design removed harmful intermediaries in the coffee system to create direct producer-consumer links. While intended to promote fairness, it also disrupted economic dependencies.	Efforts to simplify systems for sustainability can unintentionally destabilise complex relational structures.
Understanding Nonhumans through Human Understanding	Participants anthropomorphised nonhuman entities to advocate for them, using human traits like speech and land ownership. This reinforced human frameworks instead of genuinely recognising nonhuman agency.	Human-centred modes of empathy risk projecting assumptions onto nonhumans, undermining their distinct ways of being.
Designing for the Ecosystem by Prioritising the Human	Designs often began by addressing human concerns, treating ecological needs as secondary. This limited the scope of more-than-human thinking.	Humans were prioritised over ecological interdependence, revealing difficulty in decentring human perspectives.
Achieving Sustainability through More Production	Some designs proposed new infrastructures and technologies to reduce harm, yet these introduced further production and potential environmental strain.	Sustainability efforts can paradoxically rely on further consumption, perpetuating the issues they seek to resolve.

Table 7 *Paradoxes of Speculative Designs*

Paradoxes are valuable provocations for reflection in more-than-human design (Dörrenbächer et al., 2024). To support HCI researchers in attuning to more-than-human complexities, it is essential to make visible the tensions and contradictions that arise when designing beyond human-centred perspectives. By critically engaging with these tensions, HCI researchers can reflect on and develop a deeper understanding of their role within entangled more-than-human systems.

4.3 Inherent Complexities of More-Than-Human Design

In Phase 2 of the study, insights from interviews with scholars who work within the field of more-than-human design reveal the inherent complexities of more-than-human design that

HCI researchers and designers must encounter including relational entanglements, empathy gaps, power relations, and uncertainties, situatedness, and ongoing responsibilities.

4.3.1 Relational Entanglements

Tessa described a transformation in her own research—from wearable technologies to biodesign and food systems—motivated by growing concerns for sustainability. Central to this shift is her understanding that humans are not autonomous individuals, but holobionts—composite organisms formed in symbiosis with bacteria, microbes, and other life forms. Yet adopting this relational view is not always intuitive. She put it, "we are nature, and we are fundamentally entangled." This framing highlights that more-than-human design is not just about others—it is about ourselves as relational beings, embedded in living systems:

> *"As you make food with your hands, you're bringing part of you as a holobiont, you're bringing part of you into the food and you're sharing that. So, as you share your food cultures, you're also sharing your more than human cultures" (Tessa).*

This intersection between ecology and culture is further highlighted in her stories about yogurt-making practices, where microbial relationships are interwoven with histories of migration, environmental adaptation, and cultural memory—underscoring that more-than-human relationships are as much cultural and systemic as they are biological.

Nonetheless, Elias stated that sustainability often centres human futures. In more-than-human design, sustainability requires an ethical reorientation in which non-human entities are recognised as having intrinsic value and agency, rather than being instrumentalised for human benefit. One of the central challenges that emerged from the expert interviews is how to decentre the human without erasing the designer's responsibility. Alex discussed this in terms of enabling nonhuman agency to shape the design process. He explained:

> *"Decentring, for me, is something really important … to let the agency of other living beings act and transform and shape the design process" (Alex).*

Linus highlighted the role of technologies as nonhuman actors within HCI, giving an example of artificial intelligence. He described that technologies are not merely passive instruments but entities with their own forms of agency that can be engaged with rather than only used.

> *"The other agencies one works with are often technologies or things that we create" (Linus).*

However, Linus's framing of more-than-human design started from a standpoint that reasserted the boundaries between human and nonhuman. For example, he referred to more-than-human as a "negative definition"—"anything but the human"—rather than framing humans as part of a co-constituted world.

> *"What is the more than human? Because more than human is like a negative definition—it's the anything but the human" (Linus).*

This perspective reflects human exceptionalism and an anthropocentric orientation, where the human is detached from others and remains the locus of action, intention, and design agency. While Linus acknowledged the intention of decentring the human, his framing reinforced a dualistic separation between humans and nonhumans, showing that even practitioners who are experienced in the field of more than human design can struggle to practice relational thinking and to embrace the idea of humans as one entity in the systems.

4.3.2 Empathy Gaps

Tessa observed that many students from traditional, Western design backgrounds struggle to move away from egocentric and individualistic assumptions. She noted:

> *"We're trained to be egotistical and to also be individualistic... it makes it very hard for people to put themselves in another person's position, let alone the position of something that might have eyes but perceive radically differently" (Tessa).*

These deep-rooted paradigms make it difficult to imagine or perceive from nonhuman perspectives. She emphasised, "there are as many perspectives as there are entities on the planet," asserting that truth and understanding are inherently plural and perspectival, shaped by both biological and cultural dimensions. On the other hand, while nonhumans can be collaborators in design, they do not speak in human language. Elias emphasised how traditional evaluation tools like interviews—used to investigate user needs—fall short when the "users" are birds, microbes, or trees. Such work demands empathy, patience, and careful interpretation—all while recognising that the designer is still imposing meaning onto something that cannot speak back. He described the process as involving two levels: empathising, or "data gathering," followed by sense-making.

> *"You're working in kind of two levels, like data gathering or... empathising and then the kind of sense making or defining" (Elias).*

Designers must therefore both act on their behalf and with them. Alex proposed decentring the designer by taking a secondary role, disappearing altogether, or designing spaces that allow other beings to shape the outcome. This insight was paired with his critique of built environments—like heating or humidity systems—that are implicitly configured for human comfort. He suggested reimagining domestic and urban spaces as shared habitats, where design becomes a means of rebalancing power and fostering reciprocity.

> *"I think that the main challenge is that plants, rocks, animals don't talk... Our brains are wired to design from a human-centred perspective; we need to push ourselves to change that continuously" (Alex).*

4.3.3 Power Relations

Alex added another dimension by focusing on the asymmetry of agency and voice in more-than-human design. He noted:

> *"We are collaborators with them, but at the same time we are the speaking subjects... you want to involve them, but at the same time, you are interpreting and acting on their behalf" (Alex).*

Alex acknowledged the inherent ethical ambiguity: even in attempts to decentre themselves, designers remain the only speaking subjects, responsible for interpreting and enacting decisions on behalf of others. This raises difficult questions about voice, representation, and accountability. Elias and Casey both also addressed this issue of representation.

Casey described design as a form of power: it proposes change, and as such, always carries ethical responsibility. Her work focuses on the asymmetries of voice, particularly where human hierarchies and relational dynamics blur traditional boundaries. This highlights how some human groups—such as children and women—have historically been treated as "less than human," their perspectives routinely marginalised or spoken for. In the design of technologies and environments, children are often positioned as recipients of care or future adults, rather than as agents with voice and value in the present. Casey's research with children addresses this asymmetry by questioning who gets to define needs, express experience, and influence design. She points out that in many child-centred projects, adults remain the interpretive authority, making decisions on behalf of children. This asymmetry is further complicated by the fluid boundaries between entities. In these contexts, bodies, sensations, and data intermingle, and yet it is typically the adult or designer who interprets and speaks for the collective experience. Designers must confront their own positionalities and reflect on how their assumptions, perspectives, and biases shape design outcomes. Rather than striving for neutrality or omniscience, she calls for reflexive and situated ethics—a continuous practice of acknowledging where one is speaking from, and how that influences the design:

> *"I consider design as a form of power, because it is proposing a change. So whether or not a designer chooses to engage with that responsibility, I think it's really difficult not to speak for other people or beings. I think that's one of the biggest challenges" (Casey).*

4.3.4 Uncertainties, Situatedness and Ongoing Responsibilities

Elias pointed out that this shift not only changes what we design, but also how we evaluate and iterate upon our designs. Working with living systems means designers must consider slow temporalities and possibly wait years to observe how ecosystems respond to their interventions. In this view, designing in the more-than-human worlds become a durational process of sense-making and attunement, rather than a checkbox to be completed during development. He framed more-than-human design as a practice of recognising mutual interdependencies. He stressed the need to "stay with the trouble" rather than resolving it—echoing Haraway (2016)'s call to remain present in complexity rather than oversimplification. He also emphasised the need to embrace long timelines in more-than-human design:

> *"You have to kind of slowly observe how your design interacts with the natural environment, and that is a process that might take several years… that is something you just have to accept" (Elias).*

Casey highlighted making and crafting together with her target actors—children and others—as a way to attend to unfolding relationships and shared worlds. She noted:

> *"I think design is always situated… As a designer, I believe in making a mess. I feel like things are effective if they're making a mess—if things get exploded, if you see more relations" (Casey).*

Ethics in more-than-human design, therefore, were characterised as situated, relational, and continuously negotiated through lived engagements. Tessa noted that her students often approached these living systems as static problems to be solved. They are not trained to understand their ongoing responsibilities when designing with and for living elements, which makes it challenging to engage with and design for the living world. Tessa emphasised the need to understand ethics as ongoing care and responsibility, especially when working with lifeforms. She noted:

> *"There are ethical imperatives. You can't just put it in and let things die… you can't just intellectually address the problem in a rigorous way and then go away. The thing you are making is sitting in a living environment" (Tessa).*

This makes more-than-human complexities not only a conceptual issue but a practical design problem, requiring continuous negotiation among many stakeholders, both human and nonhuman.

4.4 Methods for Attuning to More-Than-Human Design

Rather than offering a fixed set of tools or phases, participants described more-than-human design as an inherently situated, experimental, and relational practice. Across interviews, methods were seen not as standardised procedures but as means of sensitisation, attunement, and relational exploration. These practices are often embodied, iterative, and context-responsive, aimed at cultivating new ways of perceiving and relating beyond human-centred norms. Central to this process is a shift in values and mindsets—an openness to rethinking dominant narratives, ethical orientations, and assumptions about agency, knowledge, and responsibility.

4.4.1 Sensory and Embodied Attunement

Both Tessa and Casey foregrounded hands-on, embodied practices as essential entry points into more-than-human design. Tessa encouraged practitioners to engage all the senses—including underutilised ones like smell, taste, and touch—in order to cultivate alternative modes of perception. She spoke of "walking with," lying in gardens, and baking alongside others as examples of how designers can immerse themselves in relational and sensory environments, expanding beyond visual and rational cognition. The responses include:

> *"There's a concept of walking with… putting our bodies in these situations can help us perceive things differently" (Tessa),*
>
> *"Perceiving things differently is what we need to do if we want to try to sensitise ourselves to other ways of knowing, being, or doing." (Casey).*

Casey also described making and crafting as powerful means of inquiry. Through crafts with children—building bird feeders, painting with flowers, and collecting frog eggs—she explored how playful, situated engagement fosters relational awareness. For both Tessa and Casey, these activities are not about arriving at defined outcomes but about staying with the process and using material engagement as a form of knowledge generation.

Tessa, Elias and Alex shared sensitising practices rooted in ethnography and cohabitation. Tessa suggested engaging in sensory ethnographic studies within the environment to sensitise oneself and attune to relational and more-than-human ways of thinking. Elias utilised sensing technologies and critical reflection prompts to complement experiential observations, bringing a blended empirical and reflective sensibility to his work. He suggested participatory ethnography tailored to specific contexts and encouraged "noticing" as a central practice: a deliberate act of attunement to ecological systems and interactions that might otherwise be ignored in conventional design:

> *"Noticing is the idea of… really seeing details in the environment to understand how different aspects are connected and to maybe see beyond the most human-centred perspective. So, there you could use kind of focusing exercises. I mean, it's a bit like meditations." (Elias).*
>
> *"You can also complement that kind of sensing with your own senses by using computational technologies. So, for example, moisture sensors, carbon dioxide sensors, basically anything that can kind of give data about the environment" (Elias).*

His toolkit included multi-species personas, and system mapping, adapted from traditional human-centred methods to include nonhuman entities. These techniques helped participants visualise interdependencies and roles within multispecies ecologies.

> *"I would recommend more than human systems mapping, because the whole point with more than human design is this kind of interdependence, mutual relationships, and so on, and those relationships you can kind of visualise by mapping systems" (Elias).*

Alex extended these insights through multispecies autoethnography—a method that draws from lived, first-person engagements with nonhuman others. His approach involved cohabitation with plants, daily shared experiences that challenge assumptions about resource use, territory, and presence. He guided students through relational sketching, writing letters to nonhuman beings, sensory journaling, and ethnographies that decentre the human

perspective. Importantly, Alex emphasised that these practices should be maintained throughout the design process, not just during early research or ideation. He argued becoming sensitised and attuned to more-than-human entanglements is a continuous, embodied effort requiring repeated confrontation with the embedded habits of human-centred design. He insisted that attunement cannot be taught abstractly; it must be lived, embodied, and personal:

> *"I cohabit with plants… sharing space, water, heating… they are another inhabitant of the place… sketching human and non-human relations… doing sensory ethnography, writing letters, diary studies" (Alex).*

4.4.2 Participatory Engagements

Tessa conducts participatory research through design workshops, bringing together diverse participants to collaboratively address conflicts through food-related activities, as human can relate to food.

> *"Food just felt really like a fantastic trigger point… it engages all of our senses… So, I use it as a tangible tool in design, but I also look at systemic change… One of my key methodologies is workshops… I try and bring communities and public authorities together" (Tessa).*

Linus sees artificial intelligent agents as active participants to explore who holds the control, who is usually getting ignored which might be agencies of other entities.

> *"… Working with, say Chat GPT or something like that or another sort of AI based tool… to really pay attention to something that's normally ignored and that might be the agencies of other entities" (Linus).*

4.4.3 Embracing Situated Experiments and Emergences

Linus resisted offering fixed methods and emphasised the experimental and emergent practice, arguing that methods for more-than-human design are emergent. He characterised the field as still experimental, where methodological insights come from trying things out—be it through speculative scenarios, future-oriented design, or AI agents that challenge conventional design interactions. He stated:

> *"It's almost like doing the design and developing the tools are done simultaneously… We've tried things. I mean, we're trying to experiment" (Linus).*

For Linus, these experiments are not about refining techniques but about reorienting thought and practice, often unsettling the familiar to open up new relational questions. This view aligns with Casey's perspective that there is no single essential tool or phase in the design process that guarantees more-than-human thinking. Like Linus, she stressed that design is always situated, shaped by the unique constellation of materials, people, places, and power dynamics.

Therefore, the appropriateness of a method depends not only on the stage of design, but on the ethical and political stakes of the context itself.

> *"I think design is always situated... There might be a certain design situation that makes sense to me to make differently or to make in one way or to use a different kind of method. I'm trying to think of an example. I can't think of one" (Casey).*

4.5 Attunement Efforts to More-Than-Human Worlds

As part of the reflective research process, the researchers individually engaged in attunement practices to explore the complexities of more-than-human design from a first-person perspective. These activities were not designed to produce generalisable knowledge, but rather to sensitise and attune the researchers themselves—inviting critical reflection on their assumptions, positionalities, and capacities for noticing more-than-human entanglements in everyday contexts (Figure 4).

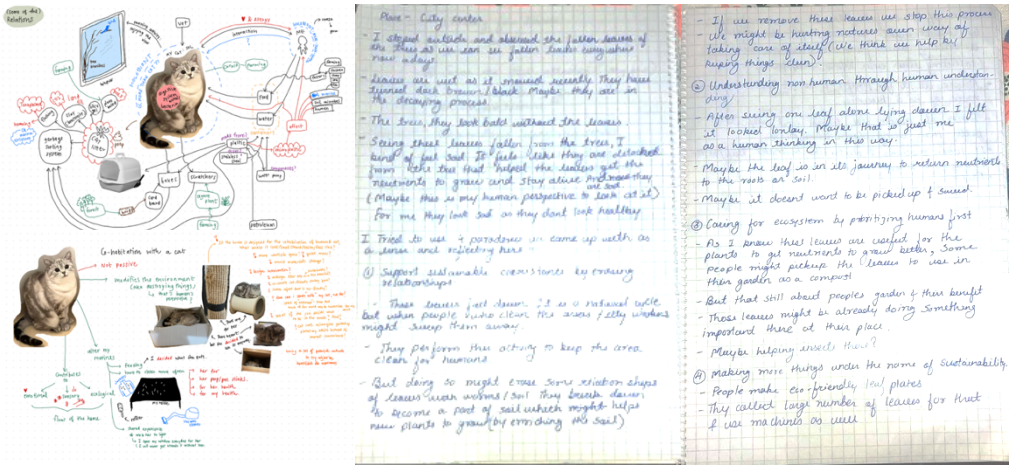


Figure 4 *Example of Attunement Practice Outcomes*

4.5.1 Persistent Anthropocentric Tendencies

Both activities revealed the difficulty of decentring the human perspective, even when intentionally attempting to do so. In one case, while observing a leave and trying to understand its ecological and relational significance, one of us found herself projecting human emotions such as loneliness onto the leaves. These emotional assumptions reflected a persistent anthropocentric habit of interpreting nonhuman entities through human categories and metaphors. Another described the cat's actions in influencing and modifying the space as "destroying things," which reflects a human perspective on the behaviour of other entities. Although well-intentioned, these perspectives risk oversimplifying and misinterpreting more-than-human worlds. Recognising this, we became aware of how deeply human perspectives can be embedded in perception—even in moments of reflection.

Therefore, alongside the attempt to attune oneself to these complexities, what also became significant during the reflections were the aspects that could not be accessed through human perception—moments and motivations that remained untranslatable. Some nonhuman entities resisted interpretation, prompting those trying to understand to revert to human frameworks—assuming intent, attributing emotions, or interpreting behaviour through a

human logic of care. This realisation led to an understanding of these gaps not as failures, but as reminders that not all experiences are available for human comprehension.

4.5.2 Attuning to the Invisible

Through the attunement practices—including observation, cohabitation, and the exploration of dynamic interactions, negotiations, and boundaries—dimensions that are often hidden or overlooked became more visible. For example, cohabitation with a domestic cat illuminated previously unnoticed patterns of co-dependence, spatial sharing and negotiation, and affective exchange. By reflecting on the presence of fallen tree leaves through the lens of paradoxes identified in the speculative design workshop, and by mapping the systems and relationships surrounding the cat, we began to perceive the invisible—ranging from immediate environmental factors to broader ecological and socio-technical interconnections within these entangled worlds. These experiences prompted critical reflection and encouraged a rethinking of how we respond to such complexities in more attuned and responsible ways.

4.5.3 Questions as Tools for Rethinking

A central part of attunement practices was the use of questions—not to find fixed answers, but to open up perception and challenge one's own assumptions. Basic questions about entities and interactions between them brought ourselves to deeper, more uncomfortable inquiries. Questions became tools for thinking—inviting a more tentative, situated, and reflective engagement with the more-than-human. Questions can be used as tools to disrupt inherited ways of knowing. In more-than-human contexts, they provoke thought and support reflection on implicit assumptions—such as anthropocentrism, dominance, or the desire for control. Unlike answers, which often close down inquiry, questions create openings. They invite uncertainty, ambiguity, and multiple interpretations. Rather than resolving complexity, they sustained a practice of staying with it—enabling more critical, situated, and responsive forms of reflection and attunement.

5. Analysis: Towards a Scaffold

This section presents the development of our scaffold for attuning HCI researchers to the complexities of designing in entangled worlds. The study revealed that attunement is not achieved through fixed solutions but emerges through sustained, reflective, and relational engagement. It involves qualities such as wondering rather than following prescriptions, decentring the human, engaging sensory awareness, embracing uncertainty, and critically reflecting on one's assumptions and ethical responsibilities. We came to see attunement as a recursive, personal process rather than a linear path. Since many HCI researchers found more-than-human design abstract and irrelevant, the scaffold must support not just behavioural change but deeper shifts in mindset. These shifts often involve discomfort and ambiguity—conditions that can serve as openings for ethical awareness and learning.

> *"I look at the values, the narratives and the mindsets that drive out behaviours. And if we can shift our values, and if we shift can transform the stories that we're telling, we can shift our mindsets and then our behaviours would follow" (Tessa).*

Our scaffold proposed three entangled *modes of attunement as central dynamics in supporting HCI researchers' to engage with the inherent complexities in more-than-human design—Mode of Wondering, Mode of Sensing* and *Mode of Reflecting*—as the central dynamics of attunement (Figure 5). Each mode is supported by the qualities of attunement.

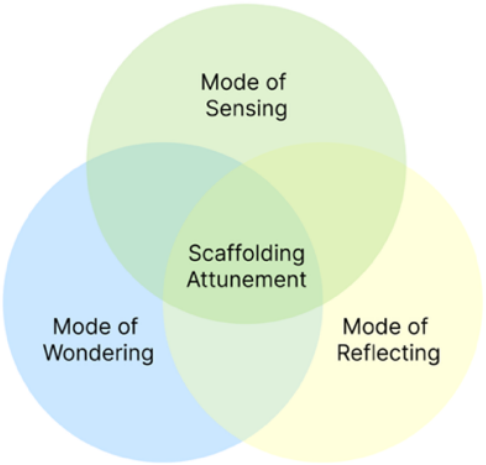


Figure 5 *Scaffold for Attuning HCI Researchers to More-Than-Human Design*

The Mode of Wondering aims to open up new ways of sensing, imagining, and approaching design situations. It encourages an exploratory, receptive, and imaginative mental stance, with a focus on the future or emergent present. This mode invites speculative and creative questioning, such as "What if?" or "How might…?", encouraging designers to engage in ideation beyond the constraints of current paradigms. Its orientation is toward speculation and imagination, and it often leads to unexpected insights, openness, and expanded perspectives:

> *"Wonder is a good word, because to wonder is to both … be amazed and in awe, but also to be curious and kind of asking questions" (Elias),*

> *"It might also be just thinking that you go into a situation not to extract what you need, but really be open for what's there" (Linus).*

The Mode of Sensing focuses on embodied and multisensory awareness. It promotes a grounded, attentive, and embodied stance, directing attention to the immediate present. Question types in this mode include "What is here?", "What do I sense?", and "What is moving?", fostering practices of slowing down, noticing, and attuning to the material and ecological world. The orientation emphasises dwelling and noticing, supporting outcomes such as grounded sensitivity, perceptual depth, and embodied awareness:

> *"There are these embodied ways of knowing that are fundamental to learning the world" (Tessa).*

The Mode of Reflecting focuses on deepening understanding of assumptions, consequences, and responsibilities in design. It encourages a critical, analytic, and ethically attentive stance, often directed toward the present and past, particularly systemic and historical dimensions. Guiding questions include "Why?", "Who benefits?", and "What's at stake?", emphasising critical examination. This mode orients toward questioning, tracing, and revealing, leading to deeper accountability, ethical awareness, and informed positioning:

> *"I think the main point … is decentring yourself" (Alex),*
>
> *"I think more of like a reflective practice" (Casey).*

Importantly, these modes are not intended as linear steps, but as personal, entangled, and recursive. HCI researchers are encouraged to move fluidly between these three modes to attune themselves to the complexities of designing in entangled worlds.

Overall, this thesis explored the challenges HCI researchers face/identify when engaging with more-than-human design. Through a three-phase reflective study—including a speculative workshop, expert interviews, and researcher attunement—it identified key challenges and revealed productive tensions that exposed ethical and relational dilemmas. Based on these insights, we proposed a scaffold to support more relational, reflexive, and situated design practices.

6. Discussion

This discussion reflects on the study's findings and their contribution to more-than-human design and HCI. It begins by outlining the challenges and opportunities involved in scaffolding HCI researchers' attunement. The discussion then situates these insights within broader HCI conversations on ethical responsibility, reflexivity, and the role of researchers as world-makers—reframing questions as generative tools for critical, situated engagement.

6.1 Scaffolding HCI Researchers' Attunement

As digital technologies become entangled with social, ecological, and material worlds, design must move beyond human-cantered views toward more-than-human relationality. This shift challenges traditional HCI practice, calling for new pathways that help researchers embrace complexity and shape design that accounts for diverse human and non-human actors. This section discusses how HCI researchers can be supported in engaging with more-than-human complexities—not simply through practical tools, but through deeper, situated processes of attunement. It begins by identifying the limitations of current tools when used in isolation. It then explores how tools may instead serve as catalysts for attunement when designed to foster curiosity, embodied engagement, reflexivity, and ethical awareness. These insights underpin the scaffold proposed in this thesis, which comprises three entangled modes: Wondering, Sensing, and Reflecting.

6.1.1 Limitations of Practical Tools

Despite the growing efforts to bridge the gap between theory and practice—through the development of design guidance (Rosén et al., 2024b), and the use of tools such as tangible prompts (Buur et al., 2023) and card-based aids, which are designed to enhance creativity and facilitate idea generation, team discussions, and design processes without requiring deep domain knowledge (Wölfel and Merritt, 2013; Roy and Warren, 2019)—the speculative design workshop revealed that HCI researchers still struggled to fully engage with more-than-human thinking. Although these tools are intended to support ideation and facilitate discussions, practitioners often found them abstract and overwhelming. For this reason, close attention should also be given to researchers' relationships with the tools they use to engage with more-than-human thinking in practice (Goodman, Stolterman and Wakkary, 2011; Stolterman and Pierce, 2012).

Additionally, moments of resistance, confusion, and the persistence of anthropocentric assumptions that continued to shape practitioners' design outcomes in the workshop also underscored another critical insight—that tools, while useful, are insufficient on their own to meet the complex demands of more-than-human design. Their struggles illustrated a deeper issue: the challenge is not merely methodological but fundamentally tied to researchers' own struggles in making sense of and internalising more-than-human perspectives. Moreover, the findings suggest that the difficulty is not rooted in a lack of intention, but rather from notion that moving beyond anthropocentric thinking is not just a matter of practice—it is a personal, cognitive, reflexive and affective process of shifting one's mindset.

6.1.2 Tools for Wondering, Sensing, and Reflecting

Such challenges supported the findings that engaging with more-than-human design is not just a matter of using the right tools or following step-by-step guides or correct process. It is an ongoing, personal, and situated transformation—one that requires cultivating new dispositions, sensitivities, and attunement. Attunement, in this context, refers to the process of becoming more perceptive, sensitive, responsive, and ethically engaged with the complex relations that shape more-than-human worlds. It involves slowing down, noticing differently, and situating one's actions within entangled ecosystems of care, responsibility, and consequence.

The inherent complexities of more-than-human design are unavoidable. HCI researchers can benefit from the three modes of attunement (Table 8) and from tools with qualities that support attunement, helping them engage with these complexities more thoughtfully. To enter the modes of attunement, researchers may need support from tools and methods that enable them to be curious, attuned and critical with more-than-human complexities. These tools often operate between modes or activate several modes of attunement simultaneously. This fluidity reflects the recursive, entangled nature of attunement itself.

Categories	Mode of Wondering	Mode of Sensing	Mode of Reflecting
Purpose	To open up new ways of sensing (e.g. seeing, feeling), approaching and imagining	To attune through embodied, multisensory awareness	To deepen understanding of assumptions, consequences, and responsibilities
Mental Stance	Exploratory, receptive, imaginative	Grounded, attentive, embodied	Critical, analytic, ethically attentive
Focus	Future, emergent present	Immediate present	Present and past
Question Types	What if / How might / What else / Could this be…?	What is here / What do I sense / What is moving / What resists…?	Why / Who benefits / Whose view / What's at stake…?
Orientation	Speculation, creative ideation, imagination	Slow down, dwell, attune, notice	Reflect, question, trace, reveal
Outcomes	Openness, unexpected insights, expanded perspective	Embodied awareness, grounded sensitivity, perceptual depth	Deeper accountability, ethics, informed positioning

Table 8 *Modes of Attunement*

As the mode of Sensing foregrounds embodied and multisensory awareness, our own bodies can serve as powerful tools for attunement. Empathy is a crucial element in the design process. Yet, gaps often emerge when human researchers attempt to empathise with nonhuman entities. Despite the attempt to empathise, there is a tendency for them to revert to human ways of knowing and anthropocentric assumptions—often leading to the humanisation of these actors (Dörrenbächer et al., 2024). Empathising with entities that do not communicate in the same way in entangled more-than-human worlds requires more than conversations but embodied attunement. Many tools focus on this situated and experiential nature, including ethnographic and participatory methods such as "walking with," cohabitation practices, participatory observation, and simply spending time in the environment, interactions, or with the entities being designed with or for. Such approach aligns with more-than-human frameworks that emphasise shared presence, multisensory learning and embeddedness (Forlano, 2017; Cerna et al., 2024; Westerlaken and Sandelin, 2024; Núñez-Pacheco and Rosén, 2024). Though attunement through lived, multisensory encounters can help HCI researchers cultivate a more situated understanding of more-than-human relations. However, it is important to recognise that some things may remain beyond human understanding—and that is okay.

Tools that support noticing—including sketching, diary studied, reflective prompts, and close observation—help researchers become aware of the dynamics and relationships that are often overlooked, encouraging reflective engagement (Rosén, 2022). Yet, noticing should not be merely about being aware of what surrounds the designer, but also of recognising one's own positionality and the ethical implications of how, what, and why one perceives and responds. This way, noticing becomes an invitation to reflexivity and positional awareness to continuously question one's own situatedness, assumptions, and responsibilities. Some tools aim to make complexities visible. One of them is through systemic representations, such as system maps, species profile cards, and multi-species personas. These tools help researchers

visualise interconnections, relational needs, and entangled agencies across human and nonhuman actors (Rosén et al., 2024b). They offer structured but adaptive ways of exploring relationships and reflect the influence of system thinking (Capra, 1996; Nelson and Stolterman, 2014) and relational ontologies (Sidorkin, 2002; Frauenberger, 2019)

Hands-on making, participatory workshops, and collaborative ethnographic methods—involving both human and nonhuman participants—were also identified as valuable tools for attunement. These approaches support material dialogue, co-creation, and shared inquiry, where making is not limited to functional prototyping, but become a way of enegaing with entities, materials, and systems. In doing so, these tools generate emergent knowledge while embracing uncertainty, experimentation, and situatedness in design (Croon, 2022; Epp et al., 2024; Núñez-Pacheco and Rosén, 2024). Moreover, participatory and multispecies methods also encourage the decentring of human, recognising nohuman as co-participants rather than passive subjects in the design process (Westerlaken and Sandelin, 2024; Cerna et al., 2024; Wilde and Lenskjold, 2024). These tools help researchers expand their ethical scope, challenging anthropocentric assumptions and fostering more inclusive forms of understanding and care (Dörrenbächer et al., 2024).

Narrative and speculative approaches—such as storytelling, metaphor, and speculative fiction—also emerged as powerful tools for wondering, sensing, and reflecting. These methods invite HCI researchers into experimental and imaginative engagement with entangled presents and potential futures (Laurien et al., 2024), rather than offering prescriptive answers, opening space for relational and ethical exploration through stories, fictions, and metaphors. Several tools are explicitly designed to provoke ethical reflection and critical thinking by offering prompts and questions that challenge researchers to rethink their own assumptions (Rosén et al., 2024b).

However, these tools are neither mandatory nor universal solutions for attunement. They are offered as supports for the slow, personal, and transformative work of caring, noticing, and responding differently. Researchers should explore them and aim to further challenge themselves to reconsider their assumptions and to shift from behaviour change to deeper ethical and relational awareness. These tools do not function in isolation, but often work across the interwoven modes of Wondering, Sensing, and Reflecting—modes that together form our scaffold proposed in this thesis. Our scaffold is not a prescriptive framework, but a situated, flexible support structure that encourages HCI researchers to cultivate new sensitivities, question assumptions, and develop more ethical and relational ways of designing. It aims to scaffold the slow, transformative work of attunement—making space for not only behavioural change but deeper shifts in mindset and design practice.

6.2 More-Than-Human Design as Thoughtful Practices

Design and making always occur within larger, interconnected systems (Capra, 1996; Nelson and Stolterman, 2014). Design practice today faces the challenge of designing with complexities, uncertainties, and entanglement in more-than-human worlds. Technologies become entangled with and influence other entities, including us, just as much as we influence them. Designers do not merely create tools or experiences; they participate in world-making—shaping relationships, values, and ways of being. In more-than-human contexts, this world-

making carries deep ethical responsibility. The transformation from anthropocentrism to more-than-human perspectives in HCI and design reflects an ethical and epistemological shift. This shift is not merely about expanding who or what is included in the design process but about rethinking the nature of design itself (Frauenberger, 2019; Giaccardi and Redström, 2020; Coşkun et al., 2022; Giaccardi, Redström and Nicenboim, 2025). The responses from the study pointed toward the importance of decentring the human in design and designing. HCI researchers and designers are among the most important actors to focus on, as they hold the power to effect change—both positively and negatively. Although it may seem paradoxical to decentre humans by centring them, it is precisely those who centre themselves who must first take the step to decentre—to begin recognising the interrelations between all entities and understand that humans are not alone in this entangled world. This process of decentring is not without discomfort. It often involves confronting one's own assumptions and learning to stay with complexities. When engaging with such complexities and unfamiliar paradigms, a reflective practice grounded in both craft and critique—combining personal intuition, experience, and ethical judgment—was called for (Agre, 2014; Löwgren and Stolterman, 2007; Goodwill, Bendor and van der Bijl-Brouwer, 2021).

Importantly, it is a personal process. When designing in the living worlds, ongoing responsibilities are unavoidable. HCI researchers and designers should cultivate a critical and reflective mindset—one that goes beyond behavioural change to address the underlying attitudes and values that shape their decisions and actions. This enables them to become more sensitive and attuned to the complexities that must be embraced (Croon, 2022; Epp, 2024). A critical, reflective and accepting mindset is essential for engaging with the proposed scaffold—that is, the modes of attunement. It is necessary to be critical and reflective in order to wonder, to sense (attentively or differently), and to reflect on one's power, decisions, and practices. In response to the ongoing responsibilities of researchers and designers, attunement must be practiced not only throughout the design process, but as a continuous part of one's life as a situated and powerful actor. This supports the view that more-than-human design must be grounded in reflexivity—an ongoing practice of acknowledging one's assumptions, biases, and positionality—which can ultimately contribute to the reframing of sustainabilities, representation and inclusivity, perceptions of technologies, and relationships between humans and nonhuman bodies in interaction design (Eriksson et al., 2024).

Through the speculative design workshop, practitioners' struggles were revealed through paradoxes embedded in their design proposals—contradictions between intentions, actions, and imagined futures. Some well-intended actions or designs can lead to dysfunctional care unintentionally (Vallgårda, 2024). These struggles indicate the challenges of overcoming anthropocentric assumptions and relational thinking (Dörrenbächer et al., 2024). However, these paradoxes or tensions emerged make visible the ethical, ontological, and relational dilemmas HCI researchers must face (Dörrenbächer et al., 2024; Núñez-Pacheco and Rosén, 2024). They serve as situated expressions of the wicked problems inherent in more-than-human design. In this sense, more-than-human design is already a challenge itself. These moments of struggle revealed openings for deeper reflection, perceiving complexities and uncertainties not as barriers but as generative design spaces—inviting attentiveness, ethical questioning, and embodied engagement (Croon, 2022; Epp et al., 2024). Our contribution

echoes the call for design practice that is situated, reflexive, and responsive to more-than-human contexts.

7. Conclusion

This thesis does not conclude with definite answers—but with even more questions. Rather than representing failure or ambiguity, this multiplication of questions is both intentional and necessary. It reflects the nature of design as a practice grounded not in solutions, but in ongoing inquiry, attunement, reflection, and ethical engagement. Throughout the study, questions consistently emerged as tools for attunement. Meaningful engagement was rarely driven by fixed methods but by questions—some uncomfortable, many unanswerable. These questions disrupt habitual ways of thinking, reveal ethical tensions, and create openings for relational understanding. They acted as invitations to wonder, sense, and reflect, to stay with uncertainties and to reframe assumptions. As more-than-human elements—including us, technologies, and others—become entangled with design practice, whether we notice it or not, design practice reveals itself as a question-driving practice: one that relies not only on efficiency and mastery, but also on sensitivity, attunement and responsiveness.

Within this thesis, a set of prompting questions emerged to support attunement across the three modes: Wondering, Sensing, and Reflecting (Table 9).

Modes	**Prompting Questions for Attunement**
Wondering	• What are the limitations of human-centred design? • What if I listened (noticed) differently? • What if I followed, instead of led, in this design process? • What care or agency goes unnoticed because it's not expressed in human terms? • How can I better understand the context or environment where a design will be used?
Sensing	• What do I notice when I begin using senses I don't normally rely on? • How does this environment feel—physically, emotionally, atmospherically? • How do my own movements and presence affect this space or its inhabitants? • If I couldn't use words, how might I relate to this entity?
Reflecting	*On relational complexities* • Which relationships come into focus, and which were overlooked? • Who or what is affected by this design beyond the immediate user? • How can the design process honour the interconnectedness of all entities involved? *On positionality and power* • How does my design exert power—and over whom or what? • Whose voices am I amplifying—and whose are left out? • Am I speaking for others—or making space to listen and respond with others? *On the paradoxes* • Am I *supporting sustainable coexistence by erasing relationships?* • Am I *understanding nonhumans through human understanding?* • Am I *designing for the ecosystem by prioritising the human?* • Am I *achieving sustainability through more production?*

Table 9 *Prompting Questions for Each Mode*

Asking questions becomes a method of design to support critical thinking, reflexivity, relational awareness, and situated ethics. That this thesis ends with more questions than it began with is, therefore, not a shortcoming but a sign of attunement in action: a reflection of the entangled, unfinished nature of designing in and with living systems. The modes of Wondering, Sensing, and Reflecting proposed in this thesis are not tools for guaranteed success, but scaffolds to support HCI researchers in staying present, critically reflecting on their past, and wondering about possible futures within the complex, shifting, and often unknowable realities of more-than-human entanglements. Ultimately, we encourage researchers to view not-knowing as a productive orientation—to see questions not as obstacles, but as guides. It is through questions that we begin to reorient design practice towards more critical, reflexive, situated, and attuned ways of being in the world.

Our thesis proposes a scaffold to support HCI researchers in attuning to the complexities of more-than-human design. Through speculative design workshops, expert interviews, and reflective attunement activities, we explored the challenges HCI researchers face/identify when designing in more-than-human entangled worlds. In response, we introduce a scaffold composed of three interwoven modes of engagement: wondering, sensing, and reflecting. These modes are not linear steps but recursive, overlapping, and situated engagements that encourage HCI researchers to open up to curiosity and unfamiliarity (wondering), slow down and attend to embodied and environmental presences (sensing), and critically examine their own assumptions, positionalities, and responsibilities (reflecting). By making visible the paradoxes, tensions, uncertainties, and complexities inherent in more-than-human design, the scaffold does not offer prescriptive steps but instead supports a mindset shift and relational awareness, enabling researchers to engage with more-than-human entanglements in design.

Looking ahead, the findings also raise questions about how we educate ourselves—and the next generation of HCI researchers and designers—for more-than-human worlds. How might we initiate these sensibilities and attunement early on? For novices, this means cultivating not just skills, but attunement habits and mindset—learning to continuously pause to wonder, sense, and reflect on their own approaches, decisions and positions within specific contexts. Design education should embrace pedagogies that emphasise slowness, care, for the interconnected world where humans are just a part of. Ultimately, attunement is not a one-time achievement but an ongoing, situated practice. The scaffold offered in this thesis serves as a support for cultivating a design culture that responds more consciously, responsibly, and relationally to the complex entanglements. As we continue to learn with more-than-human worlds, we must keep wondering, sensing, and reflecting—continually asking how we might respond within these complexities.

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APPENDIX

Workshop plan

Activity	Duration	Focus	Description	Tools/Materials
1. Coffee Reflection	15 minutes	Thinking relationally	Participants reflected on their most recent coffee experience and what made it possible, considering both human and nonhuman contributions.	Verbal prompt
2. Mapping Relationships	20 minutes	Mapping more-than-human systems	Participants individually created a coffee relationship system map, identifying humans, nonhumans, environments, tools, and other entities involved.	Agential cutting cards, paper, pens, pencils, erasers
3. Speculative Intervention	40 minutes	Designing for intervention	Participants designed speculative artefacts that could intervene in the coffee system to support sustainable coexistence, focusing on relational complexities.	Leverage pointer cards, paper, pens, prototyping materials
4. Group Reflection and Discussion	40 minutes	Reflecting on design and systems	Participants shared and discussed their speculative designs. Discussion was guided by prompts to encourage reflection on design decisions and more-than-human considerations.	Discussion prompts, shared presentation space
Wrap-Up	~5 minutes	Sharing takeaways	Participants offered final reflections and personal takeaways from the workshop.	

Agential Cutting Cards

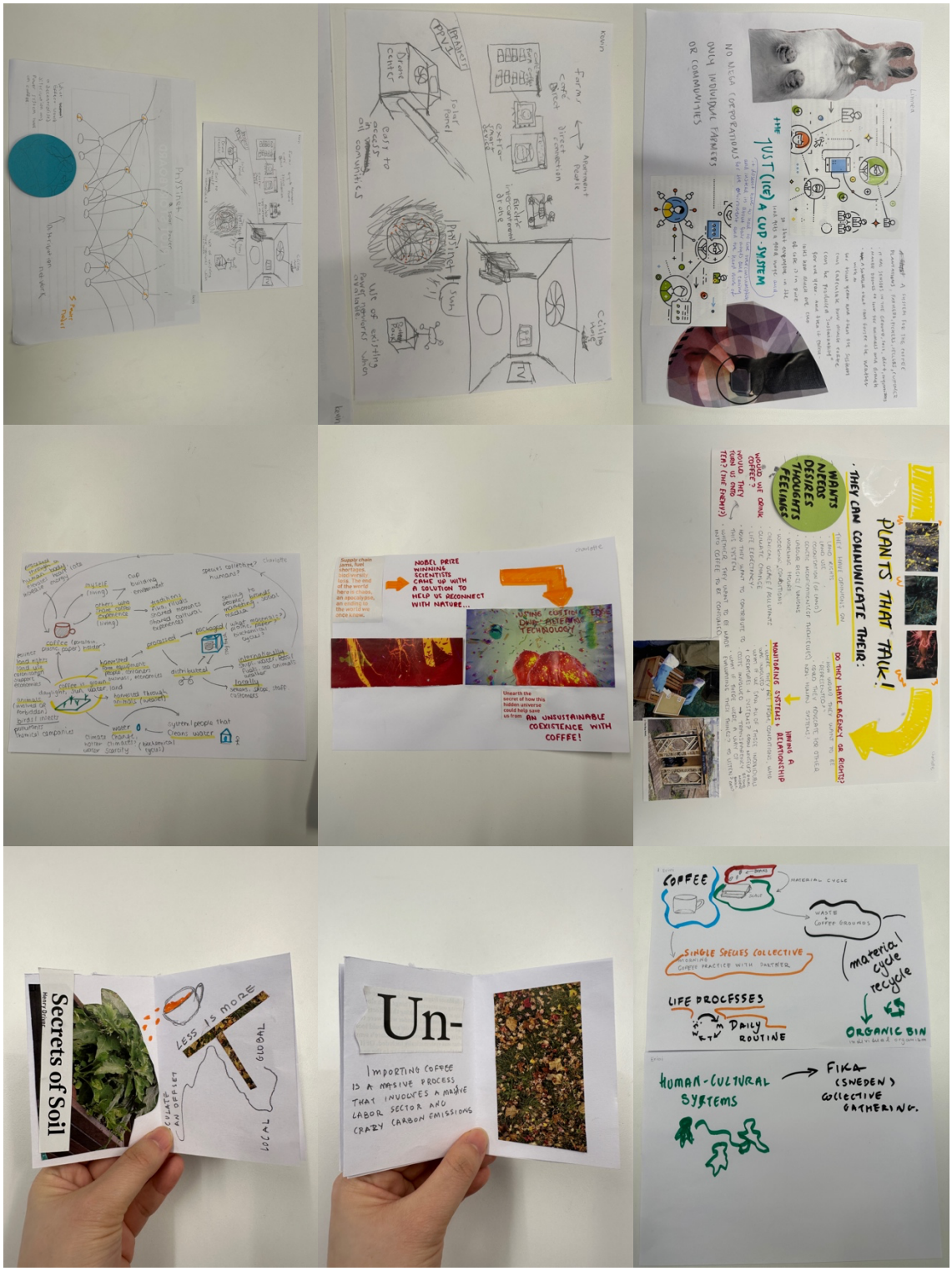
Category	Description	Examples
Individual Organism	A single living being that exists on its own but is also part of a larger system.	Animals, plants, birds, insects, fish.
Single Species Collective	A group of organisms from the same species that live together and share a way of life.	Bee hives, ant colonies, algae, moss, fungi.
Multiple Species Collective	Different species living together and helping each other in some way.	Lichen (fungi + algae), soil with microbes, compost piles, coral reefs.
Life Processes	Natural cycles that move energy and matter between living and non-living things.	Photosynthesis, breathing, decomposition, nutrient recycling.
Living System	A place where different living and non-living things interact as a connected system.	Gardens, forests, rivers, aquariums.
Biogeochemical Cycles	The movement of important elements through living and non-living parts of the planet.	Carbon cycle, nitrogen cycle, water cycle.
Atmospheric Processes	Changes in the air and climate that affect life.	Weather, seasons, global warming.
Human-Cultural Systems	Traditions, economies, and societies that shape how humans interact with nature.	Farming, religious rituals, city planning.
Tools & Technology	Objects or machines that affect how humans and non-humans interact.	Cups, computers, farming tools, AI.
Material Cycles	The movement and reuse of materials in nature and human systems.	Recycling, plastic waste breakdown, soil formation.

Leverage Pointer Cards

Front	Card Name	Description	Example
Card1	Constants, parameters, numbers (Low-Level Leverage Point)	What small adjustments (like prices, limits, or resource use) could make a difference? *These are the values that can change the outcome. Adjusting these values can be beneficial but still, bigger changes might be needed to have a long-term impact.*	A park dries out in summer. Watering plants early in the morning instead of midday reduces water loss and keeps them healthier.
Card 2	Material Stocks & Flows (Mid-Level Leverage Point)	How does the way things move through a system affect its efficiency? *The structure of the system can determine how the resources move.* *Changing the structure might take some time but have long-term benefits.*	A city collects rainwater using drains, but most of it flows away unused. If we build rain gardens and water-collecting barrels, we slow down the water flow and redirect it to be used for watering plants instead of letting it go to waste.
Card 3	Positive Feedback Loops (Mid-Level Leverage Point)	What patterns in this system reinforce themselves? How can we strengthen positive loops to create lasting change? *The positive feedback loop can amplify good or bad effects.*	Installing solar panels reduces electricity costs. Lower costs encourage more people to install solar panels, increasing clean energy use and reducing reliance on fossil fuels.
Card 4	Information Flows (Mid-Level)	What happens when people have better access to information? *Decision-making can be shaped by the available information.*	An energy company provides real-time electricity use updates to customers. Seeing how much energy they use encourages them to reduce waste.
Card 5	Goal of the System (High-Level)	What is this system trying to achieve? How could changing its goal lead to a better outcome? *Systems have their built-in goals. Changing the goals of the systems can reshape entities within the system.*	A hospital shifts its focus from treating sickness to preventing illness through public health programs, leading to better long-term outcomes.
Card 6	Mindset & Paradigm (High-Level)	What assumptions shape this system? How could a shift in thinking transform the system? *Human beliefs and assumptions influence the systems. Changing humans' mindset could lead to sustainable change?*	Designers have practiced human-centric design for a long time. When the mindset shifted to More-Than-Human design, more designs emerged that considered the needs, relationships, and perspectives of non-human entities, fostering more inclusive and sustainable systems.
Card 7	The Power to Transcend	What new ways of thinking could go beyond the current system?	Instead of designing cities for nature and animals, what if cities are part of nature? What if

	Paradigms (High-Level)	*Reimagining the existing systems allows for completely new ways of designing systems, breaking away from the current limitations*	cities aren't planned by humans alone, but co-created with ecosystems?

Workshop Outcomes



Workshop Discussion Prompts

(1) Briefly share what problems did you address and how your intervention works. What guided your decision on where to intervene in the system?
(2) Do you think your speculative artefact would truly support sustainable coexistence? How? What tensions you think might arise?
(3) What do you think designers should consider when designing for complex more-than-human systems?
(4) Were there any ethical considerations or dilemmas that emerged in your design process?
(5) How did this activity of coffee mapping influence your speculative design process?
(6) What challenges do you see in applying more-than-human design in practice?

Follow-up Interview Guide

1. Did using coffee as a lens help you engage with relational complexity in a meaningful way? If so, how? If not, what might have made it more effective? Did the familiarity of coffee make it easier or harder to recognize nonhuman relationships in the system?
2. Were there any prompt cards that significantly changed how you approached the task? Which card? How did it affect what you included or excluded in the system?
3. Did the prompt cards help you in mapping the system? Did you struggle to define what should be included in the system? What made that difficult? Did any of the cards help with this decision? If so, which one(s) and how?
4. Were there certain types of relationships (human-nonhuman, material, ecological, technological) that were harder to map or design for in the workshop? Did any card help you navigate that difficulty?
5. Were there any leverage pointer cards that made intervention choices clearer or more challenging? Which one? What about it helped (or made things difficult)?
6. Did you feel the leverage pointer cards helped in understanding the broader system dynamics? Can you recall a specific moment when a leverage card clarified a relationship or impact?
7. Did the leverage pointer cards make you rethink the ethical or relational impact of your intervention? If so, which card prompted that reflection? How did it shape your final decisions?
8. What barriers or challenges do designers face in fully engaging with more-than-human relationships? Did any card (prompt or leverage) address these challenges meaningfully?
9. When making ethical design decisions, how do you determine what to prioritise? Did any card help you think through ethical dilemmas in the system?
10. Were there any cards that felt unnecessary or unhelpful? Why do you think they didn't work for you? How could they be improved?

Expert Interview Guide

1. Would you mind starting by telling us a bit about your background and research?
2. Could you share how you define more-than-human design in your practice?

3. What kind of challenges do you think designers face when engaging with more-than-human perspectives?
4. What kind of tools or methods do you use to **sensitise yourself** to more-than-human thinking?
5. How do you use these tools in your work?
6. When working with others, how do you help them **tune into** more-than-human perspectives?
7. More-than-human design is not always intuitive for people coming from traditional design backgrounds. How do you **lure people into this space**?
8. How do you **lure yourself** into it—especially when dealing with complex, uncertain, or abstract topics?
9. What are the **key characteristics** of tools that effectively help designers understand the concept of more than human design accurately, easily, and effectively?
10. If you were to recommend **one essential tool** or **practice** for designers who are new to this, what would it be?
11. Are there any **specific exercises or activities** you use to help people engage with more-than-human systems? Are there different types of tools that work best in the **early (ideation), middle (composition), and final (implementation) phases** of the design process to help designers move beyond human perspectives and critically reflect on relational aspects of their design decisions?